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S.405 CONCRETE EXPANSION ANCHOR WORK

1. SCOPE OF WORK

- 1.1 Electrical panels, motor control centers, control panels, instruments, conduits, cable tray hangers, minor field routed pipe hangers, small tanks, galleries, ladders, stairs and other minor non vibratory equipment, structures and piping mounted on concrete floors and walls may be fastened with the concrete expansion anchor suitable for the site's environmental conditions and shall be designed, furnished and installed by the Bidder.
- 1.2 Bidder's drawings shall clearly indicate all concrete expansion anchors used in this work.
- 1.3 Expansion anchors shall not be used to fasten vibratory and fatigue loading equipments.

2. DEFINITIONS

For all expansion anchors, the following terms, when used in this specification or on the Bidder's design drawings, shall have the following meanings:

2.1 Sleeve Type Expansion Anchor

A split expansion shield, and a threaded stud bolt with cone expander, as show in manufacture's stand drawing.

2.2 Wedge Type Expansion Anchor

A split expansion ring (or a separate expansion wedge pair) and a threaded stud bolt with integral cone expander (or taper expander), as shown in manufacturer's standard drawing.

3. MATERIAL

3.1 Type And Finish Of Materials

Concrete expansion anchors and accessories shall be made of carbon steel with zinc plated finish.

Stainless steel shall be used in place of zinc coated carbon steel for outdoor application.

3.2 Strength Of Materials

The material shall have a minimum yield strength of 4220 kg/cm² for wedge type anchors, and 3515 kg/cm² for sleeve type anchors. The material shall have a minimum tensile strength of 5270 kg/cm² for all anchors.

3.3 Cross Sectional Areas

The smallest cross sectional area of the stud bolt of an expansion anchor shall not be less than the cross sectional area at the thread root.

3.4 Size Limits

The size limits for expansion anchor shall be as follows: (All sizes of bolt for sleeve and wedge type anchor shall be show on approved drawings).

3.4.1 Sleeve Type: 6 mm through 15mm diameters of stud bolts.

3.4.2 Wedge Type: 6 mm through 25 mm diameters of stud bolts.

4. INSTALLATION

4.1 General

4.1.1 All concrete expansion anchors shall be installed in accordance with the manufacturer's specifications and recommendations and requirements of this Specification. In case of conflict, the requirements of this specification shall govern. The locations where expansion anchors will be installed and the sizes of these anchors shall be as indicated on the Bidder's design drawings.

4.1.2 The minimum effective embedded length, spacing and edge distance for expansion anchors shall conform to Table S-405-1, unless otherwise indicated in clause 4.1.7 and 4.1.8 and on the standard drawings. The minimum projection of the threaded part of a bolt beyond the nut shall be 6 mm after anchor installation is completed. The projection over 6 mm shall not be removed without the consent of the Owner.

4.1.3 The overall anchor length and the hole depth required shall be determined by Bidder such that the specified minimum effective embedded length and bolt projection can be obtained.

4.1.4 Holes for the anchors may be drilled through hardened grout for finish concrete, but the effective embedded length shall be determined from the sur-

face of the rough concrete. Also, such holes may be drilled through surface repaired concrete, but the effective embedded length, anchor Clause 4.1.7 and 4.1.8. Holes shall not be drilled through grout or concrete that has not completely hardened nor shall grout or concrete be placed around un-tightened anchors.

- 4.1.5 Reuse of expansion anchors will not be permitted.
- 4.1.6 Welding on expansion anchor will not be permitted.
- 4.1.7 For anchors installed in holes drilled through surface repaired concrete, the dimension (L) given in Table S-405-1 shall be increased by the thickness of the repaired concrete cover plus the nominal diameter of the outside bar, or may be taken as 90 mm typical.
- 4.1.8 Minimum effective embedded lengths, spacing and edge distance indicated in TableS-405-1 shall be adjusted as follows under certain conditions:
 - 4.1.8.1 When two anchors of different diameters are installed adjacent to each other, the dimension of minimum spacing (“S”) shall be the average of the two “S” dimensions for these diameters respectively.
 - 4.1.8.2 If dimensions of minimum edge distance (ED), minimum edge distance to an embedded sleeve (ES) and the minimum spacing (S) cannot be maintained between two anchor assemblies, the reduced capacities of the affected anchors shall be calculated.
 - 4.1.8.3 Minimum distance from an anchor to an embedded steel plate shall equal S.
- 4.1.9 Unless otherwise indicated, no expansion anchor shall be installed at the following locations:
 - 4.1.9.1 The bottom of a concrete beam, and the top and sides of the concrete beam, at each end within a clear distance equal to $\frac{1}{4}$ clear span of the beam.
 - 4.1.9.2 The bottom of concrete slabs on metal deck, at the ribs of the deck.
 - 4.1.9.3 Masonry and hollow concrete block walls.

TABLE S-405-1
MINIMUM EFFECTIVE EMBEDDED LENGTH, SPACING AND EDGE DIS-
TANCE FOR EXPANSION ANCHORS

Nominal Bolt Diameter (mm)	Minimum Effective Embedded Length-L (mm)	Minimum Spacing-S (mm)	Minimum Edge Distance (mm)	
			ED	ES
6	25	65	85	45
10	75	115	125	65
12	100	150	175	90
16	125	190	215	110
19	150	225	250	125
25	200	300	325	165

4.2 Drilling Of Holes

- 4.2.1 For all expansion anchors, holes in concrete shall be drilled with quality carbide tripped solid bits using a rotary/percussion type power drill.
- 4.2.2 Drilling through reinforcing steel is not permitted without prior approval of the Owner.
- 4.2.3 Each hole shall be drilled precisely in order to maintain the expansion anchor's pullout value. The Tolerance on the bit shall be as specified by the anchor manufacturer for each anchor size Square or rectangular bevelled washer shall be used to maintain full bearing of the nut.
- 4.2.4 The maximum depth of the hole note be greater than the thickness of the concrete minus 64 mm.
- 4.2.5 Holes for the concrete expansion anchors shall not be drilled until the concrete has been cured for a minimum of 28 days.

- 4.2.6 If drilled holes are not used, they shall be filled with drypack grout.

4.3 Tightening Of Expansion Anchors

- 4.3.1 After the anchors for a connection are driven into the holes, they shall be brought to a “hand tight” condition so that all parts of the connection are good contact with one another.
- 4.3.2 All anchors shall be tightened to the torque values recommended by the manufacturers. Bidder shall exercise caution not to over torque the anchor causing damage to the concrete or anchor.
- 4.3.3 Tightening of anchors by means of hammering the sleeves is not permitted.
- 4.3.4 A calibrated torque wrench shall be used for tightening expansion anchors. Calibration and recalibration shall be performed on the following frequency:

For a direct reading torque wrench, on a monthly basis.

For a snap-type torque wrench, on a weekly basis.

For an air driven impact wrench, on a daily basis.

4.4 Repair Of Failures

4.4.1 Concrete Failure

This shall include all cracking or spilling of the concrete in the vicinity of an installed anchor. The concrete shall be repaired in accordance with the Bidder’s concrete repairing procedure and as approved by the Owner. After the concrete has been repaired, the anchor hole may be redrilled, in accordance with Clause 4.1.4.

4.4.2 Anchor Failure

This shall include anchor breakage, visible slippage or loosening to the extent that the anchor cannot be tightened to the installation torque.

- 4.4.2.1 If the unacceptable anchor can be removed without damaging the surrounding concrete, the hole may be redrilled and the anchor replaced with the next larger size anchor. The effective embedded length shall conform to the requirements of Table S-405-1.
- 4.4.2.2 If the unacceptable anchor cannot be removed without damaging the surrounding concrete, the anchor location shall be moved within the tolerance given for the connection. The minimum center-to-center distance from the unacceptable anchor to the replacement hole shall be 2 times the nominal bolt diameter. If this distance exceeds the given tolerance on the placement of the expansion anchors, the Owner shall be notified before proceeding.

5. INSPECTION AND TESTING

The Bidder shall be responsible for all inspection and testing work as required or as needed, unless otherwise indicated. The owner may, during, the course of the Work, inspect the various phases of the Work at the Site, for full compliance with all requirements of this Specification and the Bidder's drawings.

Any work failing to meet the specified requirements shall be rectified or replaced by the Bidder at his expense with no cost to the Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL

- 6.1 Quality assurance/control shall conform to the requirements of part 4 Sub-section 2.7.
- 6.2 The quality control procedures and documentation shall include but not limited to the following:
 - 6.2.1 Non-destructive testing.
 - 6.2.2 Packaging and shipping.
 - 6.2.3 Test procedures.
 - 6.2.4 Installation procedures.

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